Genuine Multipartite Entanglement Analysis

```
Model
-----
Hubbard ring (1-D, PBC)

System parameters
-----------------
Sites (L)         : 4
Electrons (Ne)    : 4
Hopping (t)       : 1.0
Interaction (U)   : 1.0
Mode              : uniform
n-body order      : 3
```

```
Ground-state properties
-----------------------
E0                : -3.34084762
E1                : -3.29295138

<S2>              : 0.000000
S                 : 0.0000
<Sz>              : 0.000000
Multiplicity 2S+1 : 1.0
Spin state        : Singlet (S=0)

Double occupancy
----------------
<D> total         : 0.577987
<D> / site        : 0.144497
```

```
GME summary
-----------
Extracted n-spin RDM : n = 3

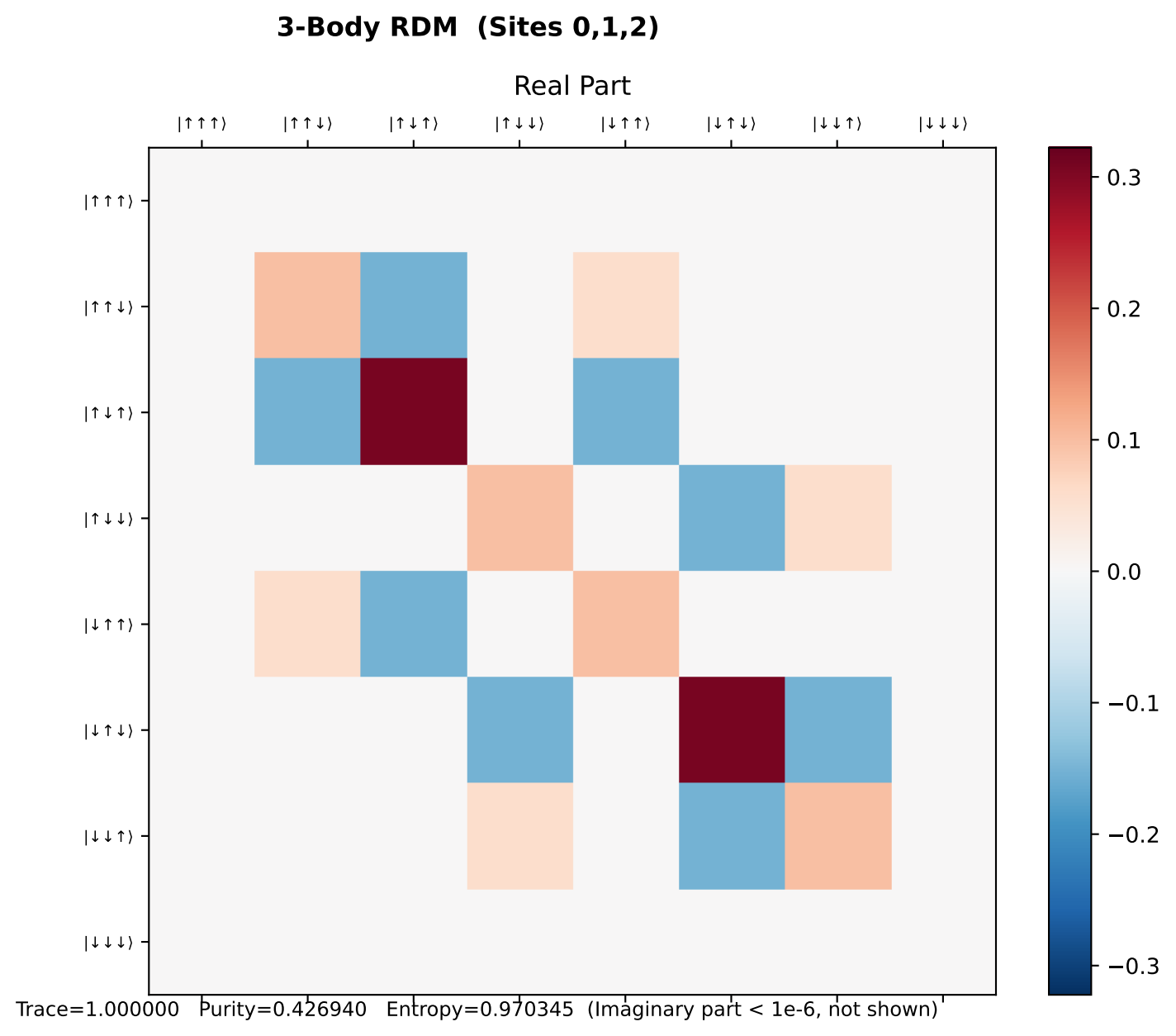
Purity:
  gamma = Tr(rho^2)

Separable cut criterion:
  gamma_A = 1  <->  rho = rho_A x rho_B

C_GME (Troiani et al.):
  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Decision logic:
  gamma=1, all Tr(rho_A^2)<1  -> GME
  gamma=1, some Tr(rho_A^2)=1 -> No GME
  gamma < 1                   -> Mixed

Date: 2026-03-20  11:58
```


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.426940 Entropy=0.970345 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4269399743
Von Neumann entropy	0.9703449365
Eigenvalue min	-2.466099e-17
Eigenvalue max	4.603212e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.42693997

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex-Roof C_GME
==========

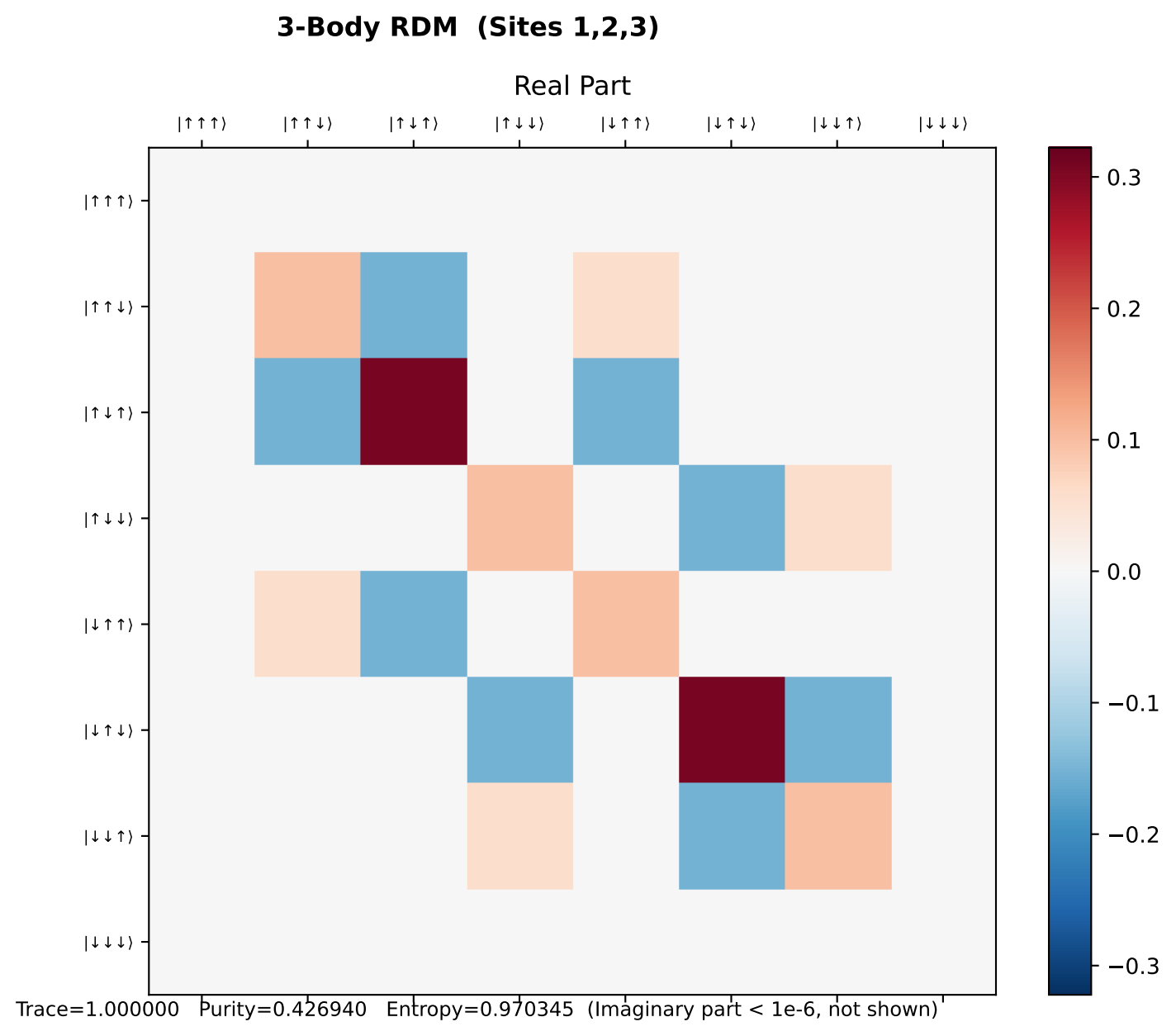
Rank(rho) = 4

C_num (convex roof) = 0.37783560 <- best
Bounds consistent : YES
Convergence : TIGHT (spread=2.5e-10)

STATUS: GME EXISTS

Convex-Roof Details
==========

Numerical (L-BFGS-B, 20 restarts):
Converged : True
Values : [0.3778, 0.3778, 0.3778, 0.3778, 0.3778, 0.3778]


3-Body RDM (Sites 1,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.426940 Entropy=0.970345 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4269399743
Von Neumann entropy	0.9703449365
Eigenvalue min	0.000000e+00
Eigenvalue max	4.603212e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.42693997

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex-Roof C_GME
==========

Rank(rho) = 4

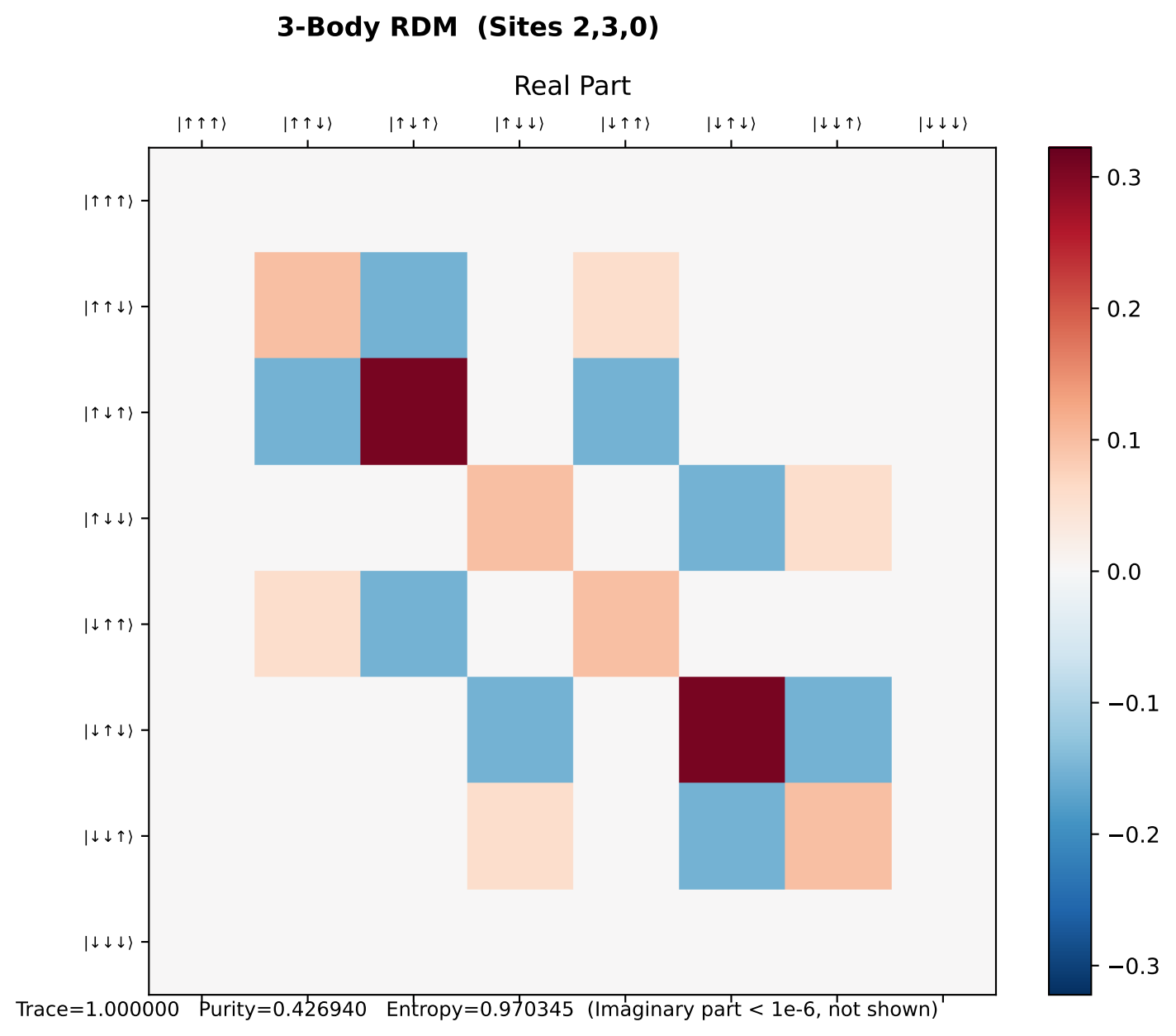
C_num (convex roof)	0.37783560 <- best
Bounds consistent	YES
Convergence	TIGHT (spread=1.1e-10)

STATUS: GME EXISTS

Convex-Roof Details
==========

Numerical (L-BFGS-B, 20 restarts):

Converged	True
Values	[0.3778, 0.3778, 0.3778, 0.3778, 0.3778, 0.3778]


3-Body RDM (Sites 2,3,0)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.426940 Entropy=0.970345 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,3,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4269399743
Von Neumann entropy	0.9703449365
Eigenvalue min	0.000000e+00
Eigenvalue max	4.603212e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.42693997

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex-Roof C_GME
==========

Rank(rho) = 4

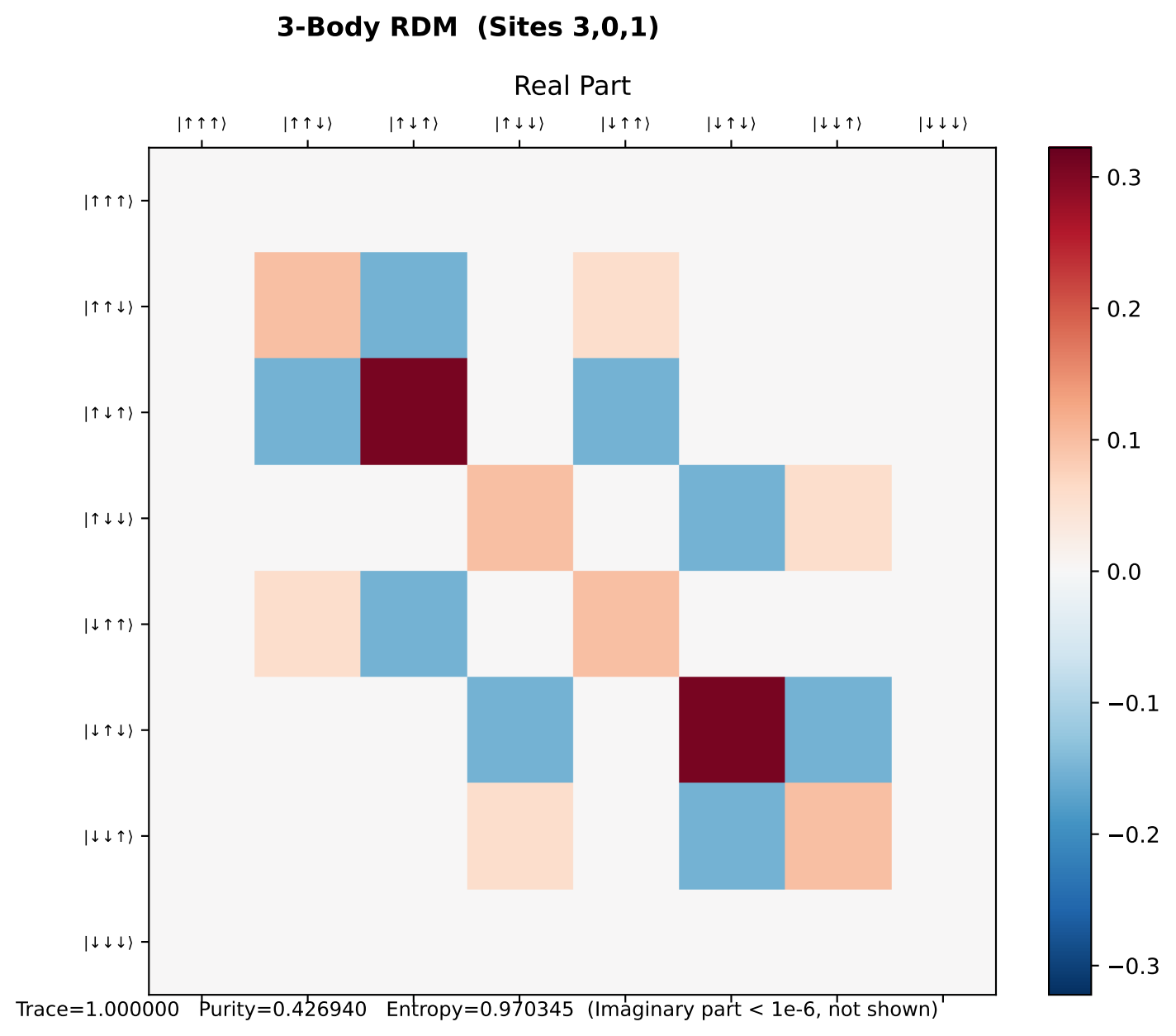
C_num (convex roof)	0.37783560 <- best
Bounds consistent	YES
Convergence	TIGHT (spread=2.2e-11)

STATUS: GME EXISTS

Convex-Roof Details
==========

Numerical (L-BFGS-B, 20 restarts):

Converged	True
Values	[0.3778, 0.3778, 0.3778, 0.3778, 0.3778, 0.3778]


3-Body RDM (Sites 3,0,1)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=0.426940 Entropy=0.970345 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 3,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4269399743
Von Neumann entropy	0.9703449365
Eigenvalue min	-7.434031e-18
Eigenvalue max	4.603212e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.42693997

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

Convex-Roof C_GME
==========

Rank(rho) = 4

C_num (convex roof) = 0.37783560 <- best
Bounds consistent : YES
Convergence : TIGHT (spread=2.4e-10)

STATUS: GME EXISTS

Convex-Roof Details
==========

Numerical (L-BFGS-B, 20 restarts):
Converged : True
Values : [0.3778, 0.3778, 0.3778, 0.3778, 0.3778, 0.3778]